

Ripple-carry adder

Adding multi-bit numbers means chaining full adders: each stage takes A, B, and carry in, and produces sum and carry out

Five plus three starter

- Starter: four-bit A equals five, B equals three, Cin zero
- Step carry from bit zero: LSBs are both one, so S0 is zero and Cout is one into bit one
- Keep stepping until all four stages reveal, sum eight, Cout zero
- Try fifteen plus one to see unsigned wrap: sum zero, Cout one
- The delay note reminds you: critical path is about N full-adder carry delays
- Wide adders use lookahead instead

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Ripple-carry adder animator

Watch carry ripple through a chain of full adders, LSB first. Each stage: $S = A \oplus B \oplus C_{in}$, $C_{out} = \text{majority}(A, B, C_{in})$. Starter: 4-bit $5 + 3 = 8$.

Starter example: 4-bit $5 + 3$. Press "Step carry $\rightarrow$ " to ripple from bit0; Cout of each FA feeds the next.

Challenge 0 / 22 cleared

Quiz: FA sum: A full adder sum bit is...

☐ $A \oplus B \oplus C_{in}$

☐ $A \& B \& C_{in}$

☐ $A | B$

☐ only $A \oplus B$

Show hint

Check

Next

Load challenge setup

Idle

Quiz: FA sum

Quiz: FA Cout

Quiz: ripple

Quiz: delay

Quiz: LSB first

Quiz: Cin

Quiz: half adder

Quiz: Cout meaning

5 + 3

15 + 1 wrap

7 + 9

Cin = 1

Long carry

Step to bit1

8-bit 100+50

8-bit overflow

0 + 0

Bit0 sum of 5+3

Quiz: generate

Quiz: propagate

Quiz: vs CLA

Reveal all

Ripple-carry adder starter

Workbook practice

- In the workbook track, draw one full adder with sum and Cout equations
- Ripple four bits for five plus three by hand
- Explain why bit one waits on bit zero's Cout
- Sketch fifteen plus one in four bits
- Name one pitfall: treating Cout as signed overflow instead of unsigned wrap

Pitfalls to watch

- Do not assume ripple is fast at sixty-four bits, carry must walk the chain
- Half adders have no C_{in} ; full adders do
- C_{in} equals one is common for two's-complement subtract
- And remember: the browser lab is literacy
- Real RTL still needs timing analysis and often carry-lookahead or prefix adders

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one ripple step with Cout feeding the next Cin
- When you are ready